```
mysql> CREATE DATABASE Hospital_Functions_Lab;

Query OK, 1 row affected (0.01 sec)


mysql> USE Hospital_Functions_Lab;

Database changed

mysql> CREATE TABLE Patients (PatientID INT PRIMARY KEY, PatientName VARCHAR(50), Age INT, Gender VARCHAR(10), WeightKg DECIMAL(5,2), ConsultationFee DECIMAL(10,2));

Query OK, 0 rows affected (0.08 sec)


mysql> INSERT INTO Patients VALUES
(201,'Asha',24,'F',55.5,600),(202,'Rohan',35,'M',72.0,800),(203,'Meera',29,'F',61.5,700),(204,'Vivek',46,'M',85.0,1000),(205,'Nitin',17,'M',58.0,500),(206,'Kavya',52,'F',68.0,900);

Query OK, 6 rows affected (0.01 sec)

Records: 6  Duplicates: 0  Warnings: 0


mysql> CREATE TABLE Medicines (MedicineID INT PRIMARY KEY, MedicineName VARCHAR(50), UnitPrice DECIMAL(10,2), Stock INT);

Query OK, 0 rows affected (0.02 sec)


mysql> INSERT INTO Medicines VALUES
(301,'Paracetamol',25,100),(302,'Amoxicillin',80,50),(303,'Cetirizine',35,75),(304,'Omeprazole',60,40),(305,'Azithromycin',120,25);

Query OK, 5 rows affected (0.01 sec)

Records: 5  Duplicates: 0  Warnings: 0


mysql> DELIMITER //

mysql> CREATE FUNCTION calculate_bmi(weight_kg DECIMAL(5,2), height_m DECIMAL(5,2))
RETURNS DECIMAL(6,2) DETERMINISTIC BEGIN RETURN weight_kg / (height_m * height_m);
END //
```

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;

mysql> SELECT calculate_bmi(55.5,1.65) AS BMI;

+-------+

| BMI |

+-------+

| 20.39 |

+-------+

1 row in set, 1 warning (0.00 sec)

mysql> DELIMITER //

mysql> CREATE FUNCTION age_category(p_age INT) RETURNS VARCHAR(10) DETERMINISTIC BEGIN IF p_age < 18 THEN RETURN 'Child'; ELSEIF p_age < 60 THEN RETURN 'Adult'; ELSE RETURN 'Senior'; END IF; END //

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;

mysql> SELECT PatientName, Age, age_category(Age) AS AgeCategory FROM Patients;

+-------------+------+-------------+

| PatientName | Age | AgeCategory |

+-------------+------+-------------+

| Asha | 24 | Adult |

| Rohan | 35 | Adult |

| Meera | 29 | Adult |

| Vivek | 46 | Adult |

| Nitin | 17 | Child |

| Kavya | 52 | Adult |

+-------------+------+-------------+

6 rows in set (0.00 sec)

mysql> DELIMITER //

mysql> CREATE FUNCTION consultation_with_tax(fee DECIMAL(10,2), tax_percent DECIMAL(5,2)) RETURNS DECIMAL(10,2) DETERMINISTIC BEGIN RETURN fee + (fee * tax_percent / 100); END //

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;

mysql> SELECT PatientName, ConsultationFee, consultation_with_tax(ConsultationFee,18) AS FeeWithTax FROM Patients;

+-------------+-----------------+------------+

| PatientName | ConsultationFee | FeeWithTax |

+-------------+-----------------+------------+

| Asha | 600.00 | 708.00 |

| Rohan | 800.00 | 944.00 |

| Meera | 700.00 | 826.00 |

| Vivek | 1000.00 | 1180.00 |

| Nitin | 500.00 | 590.00 |

| Kavya | 900.00 | 1062.00 |

+-------------+-----------------+------------+

6 rows in set (0.00 sec)

mysql> DELIMITER //

mysql> CREATE FUNCTION medicine_stock_value(unit_price DECIMAL(10,2), stock INT) RETURNS DECIMAL(12,2) DETERMINISTIC BEGIN RETURN unit_price * stock; END //

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;

```
mysql> SELECT MedicineName, UnitPrice, Stock, medicine_stock_value(UnitPrice,Stock) AS StockValue FROM Medicines;
+--------------+-----------+-------+------------+
| MedicineName | UnitPrice | Stock | StockValue |
+--------------+-----------+-------+------------+
| Paracetamol  |     25.00 |   100 |    2500.00 |
| Amoxicillin  |     80.00 |    50 |    4000.00 |
| Cetirizine   |     35.00 |    75 |    2625.00 |
| Omeprazole   |     60.00 |    40 |    2400.00 |
| Azithromycin |    120.00 |    25 |    3000.00 |
+--------------+-----------+-------+------------+
5 rows in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE patient_by_gender(IN p_gender VARCHAR(10)) BEGIN SELECT * FROM Patients WHERE Gender = p_gender; END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL patient_by_gender('F');
+-----------+-------------+------+--------+----------+-----------------+
| PatientID | PatientName | Age  | Gender | WeightKg | ConsultationFee |
+-----------+-------------+------+--------+----------+-----------------+
|       201 | Asha        |   24 | F      |    55.50 |          600.00 |
|       203 | Meera       |   29 | F      |    61.50 |          700.00 |
|       206 | Kavya       |   52 | F      |    68.00 |          900.00 |
+-----------+-------------+------+--------+----------+-----------------+
3 rows in set (0.00 sec)
```

Query OK, 0 rows affected (0.02 sec)

mysql> CALL patient_by_gender('M');

```
+-----------+-------------+------+--------+----------+-----------------+
| PatientID | PatientName | Age  | Gender | WeightKg | ConsultationFee |
+-----------+-------------+------+--------+----------+-----------------+
|       202 | Rohan       |   35 | M      |    72.00 |          800.00 |
|       204 | Vivek       |   46 | M      |    85.00 |         1000.00 |
|       205 | Nitin       |   17 | M      |    58.00 |          500.00 |
+-----------+-------------+------+--------+----------+-----------------+
```

3 rows in set (0.00 sec)

Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER //

mysql> CREATE PROCEDURE patients_above_age(IN p_age INT) BEGIN SELECT * FROM Patients WHERE Age > p_age; END //

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;

mysql> CALL patients_above_age(25);

```
+-----------+-------------+------+--------+----------+-----------------+
| PatientID | PatientName | Age  | Gender | WeightKg | ConsultationFee |
+-----------+-------------+------+--------+----------+-----------------+
|       202 | Rohan       |   35 | M      |    72.00 |          800.00 |
|       203 | Meera       |   29 | F      |    61.50 |          700.00 |
|       204 | Vivek       |   46 | M      |    85.00 |         1000.00 |
|       206 | Kavya       |   52 | F      |    68.00 |          900.00 |
```

```
+-----------+-------------+------+--------+----------+-----------------+
```

4 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

```
mysql> CALL patients_above_age(40);
+-----------+-------------+------+--------+----------+-----------------+
| PatientID | PatientName | Age  | Gender | WeightKg | ConsultationFee |
+-----------+-------------+------+--------+----------+-----------------+
|       204 | Vivek       |   46 | M      |    85.00 |         1000.00 |
|       206 | Kavya       |   52 | F      |    68.00 |          900.00 |
+-----------+-------------+------+--------+----------+-----------------+
```

2 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE update_medicine_stock(IN p_id INT, IN p_quantity INT) BEGIN
UPDATE Medicines SET Stock = Stock + p_quantity WHERE MedicineID = p_id; END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DELIMITER ;
mysql> CALL update_medicine_stock(301,20);
```

Query OK, 1 row affected (0.01 sec)

```
mysql> SELECT * FROM Medicines WHERE MedicineID = 301;
+------------+--------------+-----------+-------+
| MedicineID | MedicineName | UnitPrice | Stock |
```

```
+------------+--------------+-----------+-------+
|       301 | Paracetamol  |     25.00 |   120 |
+------------+--------------+-----------+-------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE medicine_count(OUT p_count INT) BEGIN SELECT COUNT(*) INTO p_count FROM Medicines; END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL medicine_count(@total);
Query OK, 1 row affected (0.03 sec)

mysql> SELECT @total AS TotalMedicines;
+----------------+
| TotalMedicines |
+----------------+
|              5 |
+----------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE patient_report() BEGIN SELECT PatientID, PatientName, Age, calculate_bmi(WeightKg,1.65) AS BMI, age_category(Age) AS AgeCategory FROM Patients; END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
```

```
mysql> CALL patient_report();
+-----------+-------------+------+-------+-------------+
| PatientID | PatientName | Age  | BMI   | AgeCategory |
+-----------+-------------+------+-------+-------------+
|       201 | Asha        |   24 | 20.39 | Adult       |
|       202 | Rohan       |   35 | 26.45 | Adult       |
|       203 | Meera       |   29 | 22.59 | Adult       |
|       204 | Vivek       |   46 | 31.22 | Adult       |
|       205 | Nitin       |   17 | 21.30 | Child       |
|       206 | Kavya       |   52 | 24.98 | Adult       |
+-----------+-------------+------+-------+-------------+
6 rows in set (0.00 sec)

Query OK, 0 rows affected, 6 warnings (0.03 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE medicine_report() BEGIN SELECT MedicineID, MedicineName, UnitPrice, Stock, medicine_stock_value(UnitPrice,Stock) AS StockValue, CASE WHEN Stock < 30 THEN 'Low Stock' WHEN Stock <= 70 THEN 'Medium Stock' ELSE 'High Stock' END AS StockStatus FROM Medicines; END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL medicine_report();
+------------+--------------+-----------+-------+------------+--------------+
| MedicineID | MedicineName | UnitPrice | Stock | StockValue | StockStatus  |
+------------+--------------+-----------+-------+------------+--------------+
|        301 | Paracetamol  |     25.00 |   120 |    3000.00 | High Stock   |
|        302 | Amoxicillin  |     80.00 |    50 |    4000.00 | Medium Stock |
```

| 303 | Cetirizine | 35.00 | 75 | 2625.00 | High Stock |

| 304 | Omeprazole | 60.00 | 40 | 2400.00 | Medium Stock |

| 305 | Azithromycin | 120.00 | 25 | 3000.00 | Low Stock |

+------------+--------------+-----------+-------+------------+--------------+

5 rows in set (0.00 sec)

Query OK, 0 rows affected (0.03 sec)

mysql>